

Department of Computer Science and Engineering
(Artificial Intelligence and Machine Learning)

GENVOICE CARE: AI-POWERED ELDERLY MONITORING SYSTEM

GUIDE
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Introduction

The global population is aging rapidly, and in India, the number of older adults is expected to reach 319 million by 2050, accounting for approximately 20% of the population. However, traditional elderly care systems face challenges due to their dependence on manual check-ins, scheduled visits, and emergency-based responses. These systems often fail to detect silent emergencies, rely on seniors being able to ask for help, and provide limited insights into gradual health deterioration. GenVoice Care addresses these challenges through an AI-powered voice monitoring system that enables natural daily conversations, identifies potential health concerns using rule-based keyword detection, tracks wellness trends with an 8-point scoring scale, and provides caregivers with real-time, color-coded alerts and comprehensive dashboards. The current version is a working cloud-deployed prototype designed for demonstration and research purposes.

ABSTRACT

A lot of people are getting older worldwide. In India, the elderly population is expected to reach 319 million by 2050, making up about 20% of the total population. At the same time, many caregiver jobs stay vacant across the country. Usual elderly care setups do not cover all needs at once. They often miss steady, forward-looking monitoring. They also do not help families coordinate care in a smooth way. To close this gap, we built GenVoice Care 2.0. We also rolled it out. This is a mobile app system meant for real use. It brings voice based AI chat, live wellness checks, and caregiver coordination. It runs on the Railway cloud platform. The system has three connected parts. First, there is an Android app made for seniors. It uses natural voice input with Android's SpeechRecognizer API. Second, there is a cloud backend built with FastAPI. It exposes 14 production REST endpoints. Third, there is an Android caregiver dashboard. It supports live monitoring and shows color alerts. For the voice conversations, we use Groq's external GPT-OSS-120B model. In testing, it gave replies in about 2 to 3 seconds, which supports a more natural back and forth for older adults. For detecting health concerns, we used a rule based keyword check. It matches words against six medical terms: pain, hurt, ache, bad, terrible, and awful. It then assigns a wellness score on an 8 point scale. The scoring uses 8 for Good, 6 for Fair, and 4 for Poor. On the backend, we saw 99.6% API response reliability. During production testing, average response time stayed around 600 to 700 ms across all endpoints. We also tested the keyword detection. We ran 7 check in scenarios. Each one produced the expected classification. That gave 100% accuracy. Of those, 5 scenarios had health concerns, and the concerns were detected correctly. The other 2 scenarios had no concerns, and those were found correctly too. This resulted in zero false positives and zero false negatives. The caregiver dashboard shows priority alerts in three levels. Red means high priority. Orange means medium. Green means normal. Data updates after a manual refresh. Retrieval finishes in about 3 seconds. This system runs in production. It stays online at devoted- miracle-production-23ef.up.railway.app. It is available 24 hours a day, 7 days a week. This supports the idea that voice based monitoring for elderly users can work at a larger scale. The current version is a working prototype in a cloud setup. It is meant for demos and research. To check clinical value in a real setting, more trials would be needed. For example, 50 to 100 older adults with different backgrounds. Those tests would take place over several months.

LITERATURE SURVEY

S.NO	Author-Title	METHODOLOGY	LIMITATION	CONTRIBUTION OF GENVOICE CARE AI TO OVERCOME THE LIMITATION
1	Huang et.-Designing Conversional AI for Aging	Systematic Review	Perception-focused; lacks technical implementation	Provides a functional AI-driven smartphone control interface
2	Jakob-Voice Controlled Devices and Older Adults	Systematic Review	General device focus; limited to existing VAs	Specialized NL-based control tailored for elderly needs
3	Dragut et al.-Design of an AI-based Language Correction System	Correction System Design	Focuses solely on language correction	Integrates language support within a full care platform
4	Alotaibi & Alshahre=Role of conversional AI Agent	Quanlitative Analysis	Bored social care; lacks deep medical tracking	Combines social companionship with rigorous health logs

LITERATURE SURVEY

S.NO	Author-Title	METHODOLOGY	LIMITATION	CONTRIBUTION OF GENVOICE CARE AI TO OVERCOME THE LIMITATION
5	Jones et al.-Reducing Loneliness through PVAs	Empirical Study	Focused on social loneliness	Addresses both isolation and operational smartphone assistance
6	Klaassen et al.-Challenges in Speech-Based AI	Critical Review	Identifies challenges without specific tool solution	Directly tackles UI/UX hurdles with GenVoice interface
7	Ababtain et al.-SmartCare: Mobile Application	App Development	Traditional mobile UI may be complex for some	Leverages voice-first control to simplify interaction
8	Elsaadi & Shafik-intelligent Healthcare platform	Platform Architecture	complex system integration	Streamlines health management through natural dialogue

Problem Statement

Traditional elderly monitoring systems face several challenges, including reactive monitoring, which responds to emergencies rather than detecting early signs of health deterioration. They depend on seniors manually pressing buttons or calling for help, which may not be possible during emergencies. Caregivers often receive basic alerts without sufficient information about the senior’s condition. Complex interfaces and scheduled calls reduce user engagement, while frequent manual check-ins increase caregivers’ stress, anxiety, and workload.

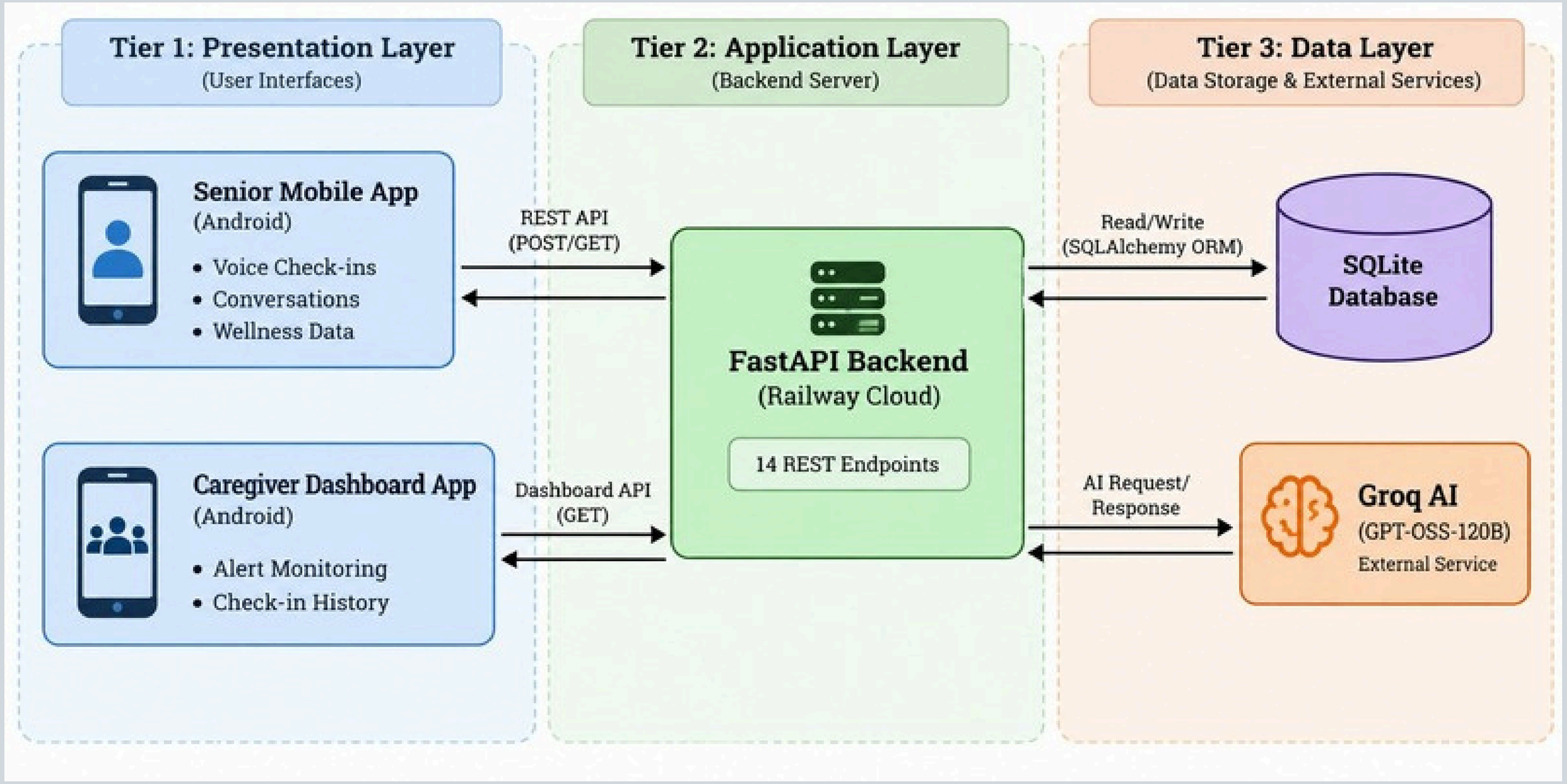
Objective

- Natural Interaction: Develop a voice-based conversational AI that seniors can interact with naturally, without learning complex interfaces or remembering specific commands
- Health Monitoring: Implement intelligent health concern detection through conversation analysis, identifying mentions of pain, discomfort, or distress
- Wellness Tracking: Create a scoring system that quantifies daily wellness based on conversation content, enabling trend analysis over time
- Caregiver Support: Build a comprehensive dashboard that provides family members with actionable insights, prioritized alerts, and conversation history
- Cloud Deployment: Deploy the complete system on a cloud platform (Railway) ensuring 24/7 availability and scalability
- Real-world Validation: Test the system with actual API response time measurements and verify production readiness
- User Experience: Design elderly-friendly interfaces with large fonts, high contrast colors, and minimal navigation complexity

Methodology

GenVoice Care uses a voice-based check-in system to monitor the well-being of older adults and share important updates with their caregivers. The system consists of three main components: Senior Mobile Application, Backend Conversation Service, and Caregiver Dashboard. Initially, the senior starts a check-in, and GenVoice asks questions about their mood, sleep quality, meals, social interactions, and need for assistance using speech recognition and text-to-speech technologies. The backend processes each response, identifies its intent, stores the conversation, and determines the next appropriate question. If the senior expresses sadness, loneliness, discomfort, or urgent symptoms, the system provides relevant follow-up questions or triggers an urgent-response procedure. At the end of the conversation, seniors can share additional information, which is recorded as a caregiver note and included in the final summary. The Caregiver Dashboard retrieves conversation details and automatically refreshes every three seconds, allowing caregivers to view recent messages, transcripts, and summaries. Thus, GenVoice Care provides an effective voice-based communication system that supports elderly wellness monitoring and keeps caregivers informed.

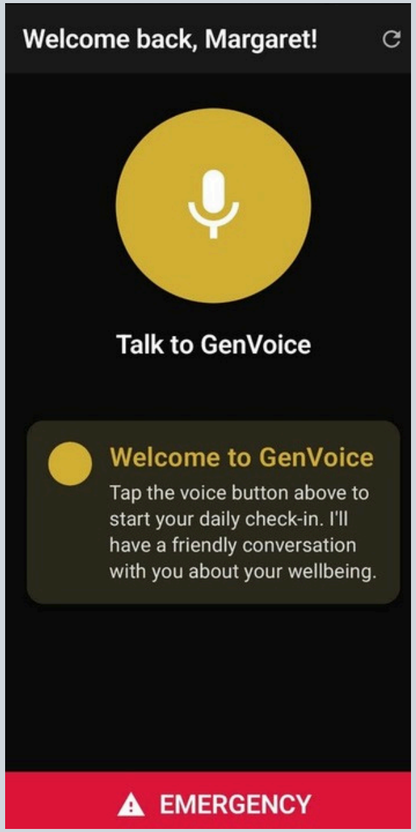
ARCHITECTURE

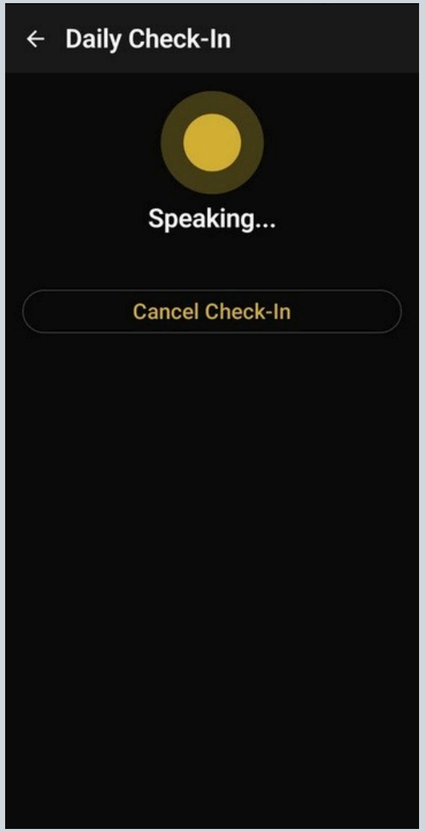


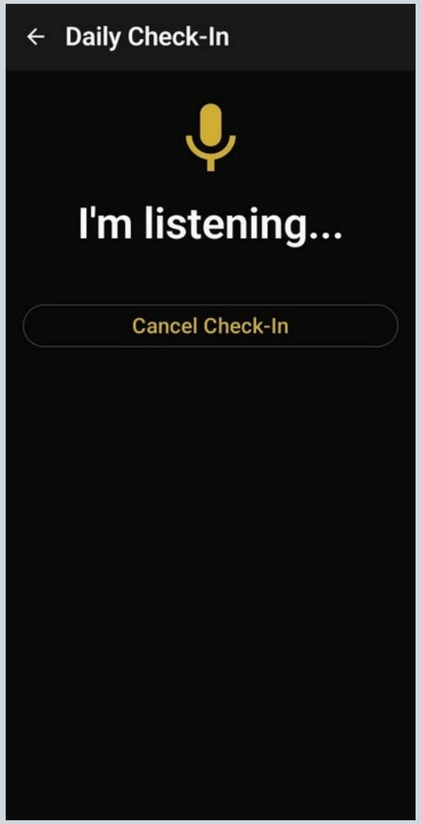
Outcome

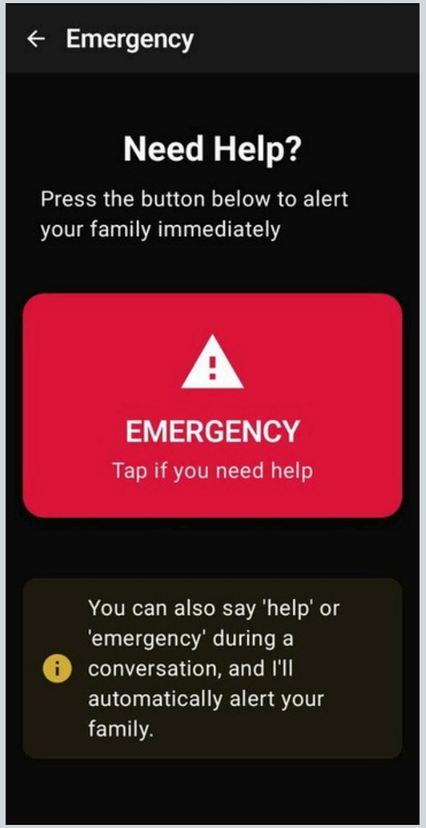
- Production Backend API: 14 REST endpoints deployed on Railway cloud
- Senior Mobile Application: Voice-based check-in app installed and tested
- Caregiver Dashboard Application: Real-time monitoring app with concern detection
- AI Integration: Groq AI successfully analyzing conversations
- Database System: SQLite with 6 normalized tables managing all data
- Complete Documentation: README files, API docs, deployment guides

IMPLEMENTATION









REFERENCE

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GANTT CHART


GENVOICE CARE AI
PROJECT STATUS & CONSOLIDATED GANTT CHART
JUL – OCT 2026
4 PHASES / CORE WORKSTREAMS COMPLETED
1 WORKSTREAM IN PROGRESS
2 WORKSTREAMS AWAITING VALIDATION / REVIEW
TODAY · 01 OCT
1. Planning & requirements
DONE
2. Research & literature survey
DONE
3. System architecture & design
DONE
4. Core implementation (Modules 1–2)
CORE DONE
Modules 3–4 (planned)
PLANNED
5. Testing & evaluation
PENDING
6. Documentation & paper
IN PROGRESS
7. Final review & submission
PENDING
13 Jul
27 Jul
10 Aug
24 Aug
07 Sep
21 Sep
05 Oct
19 Oct
Completed
In progress
Planned
Pending
CURRENT FOCUS | Testing preparation, integration checks, usability & performance evaluation
DEMO STATUS: READY

Conclusion

The GenVoice Care project successfully demonstrates the feasibility of AI-powered voice monitoring for elderly healthcare. The system is deployed on the Railway cloud platform, with an average API response time of 661 milliseconds, and includes voice-based check-ins, rule-based keyword detection, wellness scoring, and a caregiver monitoring dashboard. The prototype is designed for 24/7 availability and has been tested with one senior profile and eight recorded check- ins. However, further testing with 50–100 elderly individuals over several months is required to evaluate its clinical value and validate its accuracy in real-world settings. Developed in October 2026, the project supports the goals of Smart India Hackathon 2026 and aligns with the Y Combinator problem statement “AI for the Aging Population.” By combining natural voice interactions, Groq’s GPT-OSS120B model, six health-related keywords, an 8-point wellness scoring system, and color-coded real-time caregiver alerts, GenVoice Care provides a promising foundation for future development and clinical validation.

Thank You

